

An Alternative Approach to Locating Voltage Sag Source Side at the Point of Common Coupling Based on Power-Flow Information

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Abstract This paper proposes a new approach capable of locating the voltage sag source side in the power grid based on power-flow behavior at the common coupling point. Measuring the phase voltages and the phase currents flowing at the interested systems interconnection point, the proposed approach is a suitable solution to locate the side of symmetrical and asymmetrical voltage sag source using a set of Boolean logics. A comparative analysis is performed between the best distance relay fault locator method and the proposed approach after being assessed through a simulated case study in a Brazilian utility and industrial systems. Each simulated system has a distributed generation, active loads and power flowing at any directions. The novelty proposal is justified by the simplicity and robustness. Theoretical analysis and simulation have verified its effectiveness.

Keywords Fault location · Power flow · Power quality · Voltage sag source · Voltage sags

1 Introduction

Power quality (PQ) is important issue for both utilities and consumers (Milanovic and Gupta 2006a,b), because of the need for providing and receiving power according to the digital quality concept: continuity and compliance with power quality standards. In this context, voltage sag is the major

disturbance, as it is unavoidable and inherent in power system operation and also causes frequent interruptions in industrial processes (de Carvalho Filho et al. 2008). As a result, there have been many disagreements between utilities (transmission–distribution and distribution–distribution) and also between utilities and consumers, in the search and identification of the responsible agent for the voltage sag occurrence. Therefore, identifying the system of the source of voltage sags is crucial to support the analyses aiming to solve such issues.

A comparative evaluation of the various methods based on phasors of voltage and current measurement capable of indicating the side of voltage sag source (upstream or downstream) from the monitored bus was presented in Leborgne et al. (2006) and Polajzer et al. (2015). Among them, stand out the methods of disturbance power and energy (Parsons et al. 2000), slope of the system trajectory (Li et al. 2003), estimated resistance sign (Tayjasananant et al. 2005), real current component (Hamzah et al. 2004) and application of distance relay algorithm (Pradhan and Routray 2005). After conducting simulations of several symmetrical and asymmetrical faults in several buses of a transmission and distribution system used as a case study, it was found that the performance of the methods was much better for symmetrical faults, where 87 % of the results were correct. For asymmetrical faults, however, correct results were only 65 %. The method based on the application of the distance relay algorithm (impedance method) had the highest number of hits, 11 of 12 cases (91.6 %), proving to be an effective identifying method. In Yonghai et al. (2012), was presented an improved impedance calculation method, which uses the change rate of impedance magnitude and increment of impedance angle before and during the voltage sag event to compare with a reasonable thresholds. The improvement in Yonghai et al. (2012) was made in order to solve problems in Pradhan and Routray

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Table 1 Advantages and disadvantages of voltage sag locator phasor-based methods

Method	Advantages	Disadvantages
Disturbance power e energy	Simplicity	Not sensitive to small power-flow variations and dependent of the load type involved
Slope of the system trajectory	Threshold is not required	The method has not good performance for asymmetrical faults
Estimated resistance sign	Good performance for linear systems	The method may not be reliable for meshed systems
Real current component	Voltage sag source location for three-phase faults	Unsatisfactory performance for unbalanced faults
Impedance algorithm	Good performance for symmetrical and asymmetrical faults	Not sensitive voltage sag source at upstream side of the one power source system
Improved impedance algorithm	Sensitive voltage sag source at upstream side of one power source system	Threshold for impedance variation is required

(2005), when there is no change in the seen impedance for upstream voltage sag. Table 1 summarizes the advantages and disadvantages of various methods presented.

This paper proposes a new approach capable of identifying the side of voltage sag source (VSS) based on a set of Boolean logics capable to track the expected power-flow behavior with an good performance for symmetrical and asymmetrical faults. The comparative analysis is performed between the improved impedance method [Yonghai et al. \(2012\)](#) and the proposed approach.

The improved impedance method comes to solve the incapability presented in [Pradhan and Routray \(2005\)](#) when the voltage sag source occurs between the monitor and the unique power source of the system. The impedance calculated in (1) and (2) may does not change on this condition.

$$\Delta|Z| = \left(\frac{|Z_f|}{|Z_{pre}|} - 1 \right) \times 100 \% \quad (1)$$

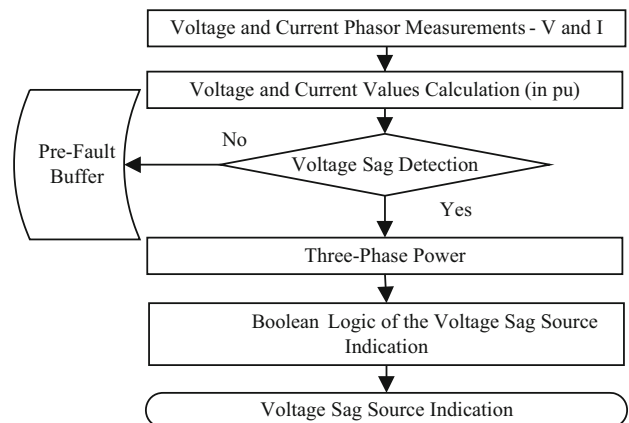
$$\Delta\angle Z = \angle Z_f - \angle Z_{pre} \quad (2)$$

where $\Delta|Z|$ delta impedance; Z_f impedance during the voltage sag; Z_{pre} impedance before the voltage sag; $\Delta\angle Z$ delta angle;

After simulations, the authors proposed two thresholds (-40% to the delta impedance and 25° to the delta angle). If $\Delta|Z| < -40\%$ and $\Delta\angle Z > 25^\circ$, then voltage sag source located at downstream side; otherwise, voltage sag source located at upstream side.

2 Locating the Source of Voltage Sags Based on Power-Flow Behavior

This approach will add a new concept to the group of ideas already proposed in previous studies, providing a simple yet efficient identifier of the side of VSS to be installed on the boundary point between two electrical systems. In an innovative way, the proposed method identifies the side of the

**Fig. 1** Flowchart of the proposed voltage sag source indication method

VSS in relation to the monitored interconnection assessing the locus of the complex power during the disturbance.

The proposed method performs, for each measurement step, through the sliding window, the flow chart shown in Fig. 1. After measuring the three voltage phasors and the three current phasors at boundary buses, in p.u., the measured values are stored as the condition before voltage sag. If voltage sag detection occurs, it obtains the three-phase active and reactive power before and during the voltage sag.

Voltage sags are detected by assessing the magnitude of voltage phasors of each phase. After the violation of the limit below 0.9 p.u. of at least one of the phasors during 0.5 cycle or more ([IEEE Recommended 2009](#); [Masetti 2010](#)), the voltage sag detection algorithm triggers the process of analyzing the locus of the complex power calculated during the voltage sag. When triggered, the proposal identifies the side of VSS based on, mostly, two concepts:

- First, the reactive power flow between two buses, separated by an impedance with prevailing inductive reactance, is primarily defined by the difference in voltage magnitude. The use of this concept [Makaliki and](#)

Leborgne (2007) is different from traditional approaches based on the active power flow as in Parsons et al. (2000), because the reactive power flow has more relevance on the proposed algorithm. When fault occurs, the reactive power flow prevails on system and this is the main reason for the voltage sags on system.

- Second, the power flow at prevoltage sag condition may be at any of the four quadrants. It is important when any kind of system is considered.

The behavior of three-phase complex power will depend on the characteristics of the systems where the intelligent electronic device (IED) is connected, such as active sources at one side only or both sides, characteristics of the loads (constant impedance, constant power or constant current) and the condition of operation at the time of the disturbance (first, second, third or fourth quadrant). The exact three-phase complex power value does not matter, but only the behavior is verified by the method as the increase or decrease in the active power flow and of the reactive power flow or the change in flow direction. The method is suitable for power sources at both sides and at only one side, including both possibilities.

For systems with power sources at both sides, the flow of reactive power always increases regardless if the fault is at the upstream or downstream side of the IED. When the reactive power increases going to the first or second quadrant, the flow is to the downstream direction, and therefore, the fault was at the downstream side. On the other hand, when the reactive power flow increases forward to the third or fourth quadrant, the fault is at the upstream side of the IED. The location of the fault is defined by the reactive power flow regardless of the active power flow.

For systems with power source at only one side, when the fault occurs in the other side, it is expected an increase in the reactive power flow as the case with source at both sides. If the fault occurs at the same side of the power source, the voltage at the other side will reduce and the load behavior will depend on the load type (constant impedance, constant power or constant current). However, it is expected a reduction in the reactive power flow to the load side (reactive consumption of the average load will reduce due to the voltage sag). The reduction in reactive power could be capacitive or inductive depending on the prefault condition. In this case, the result will be indicated by the variation in the active power flow. All possibilities of the expected behavior for any kind of system when the voltage sag source at upstream or downstream side are illustrated in Figs. 2 and 3, respectively. As example, Fig. 4 shows the power-flow behavior when a voltage sag source occurs at downstream side, considering the initial power flow at fourth quadrant.

The proposed algorithm was developed to track all possibilities illustrated in Figs. 2 and 3. It consists of a set of Boolean logics as shown in Eqs. (3–7), resulting in a feasible

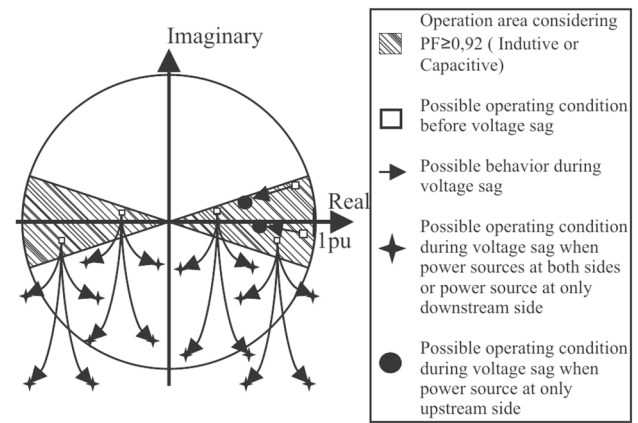


Fig. 2 Illustration of possibilities of the expected power flow behavior when voltage sag source occurred at upstream side for any kind of system

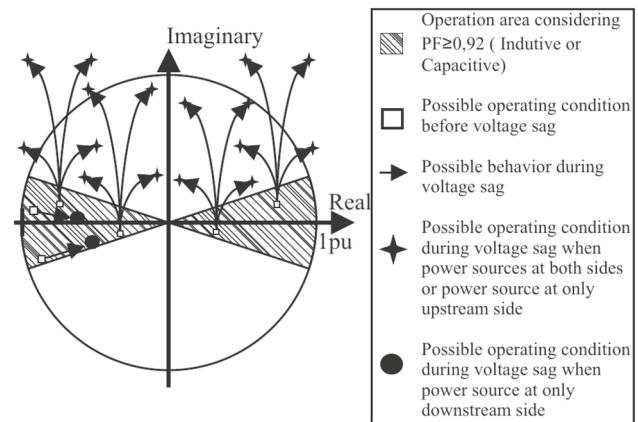


Fig. 3 Illustration of possibilities of the expected power flow behavior when voltage sag source occurred at downstream side for any kind of system

VSS indicator.

$$QD = [(Q > 0) \times (QP \geq 0)] + [(Q < 0) \times (QP \leq 0)] \quad (3)$$

$$U1 = (PP \geq 0) \times (|Q| < |QP|) \times QD \quad (4)$$

$$D1 = (PP \leq 0) \times (|Q| < |QP|) \times QD \quad (5)$$

$$U = U1 + \overline{D1} \times (\Delta Q < 0) \quad (6)$$

$$D = D1 + \overline{U1} \times (\Delta Q > 0) \quad (7)$$

where (Q) imaginary part of three-phase complex power during voltage sag; (PP , QP) real and imaginary parts of three-phase complex power on the condition of prevoltage sag, respectively; (ΔQ) variation in imaginary part of three-phase complex power from the prevoltage sag to the voltage sag condition; (QD) a Boolean variable which indicates that the reactive power did not change or there was no transition; ($U1$) first Boolean variable that indicates an upstream fault. It represents the condition of a single source system

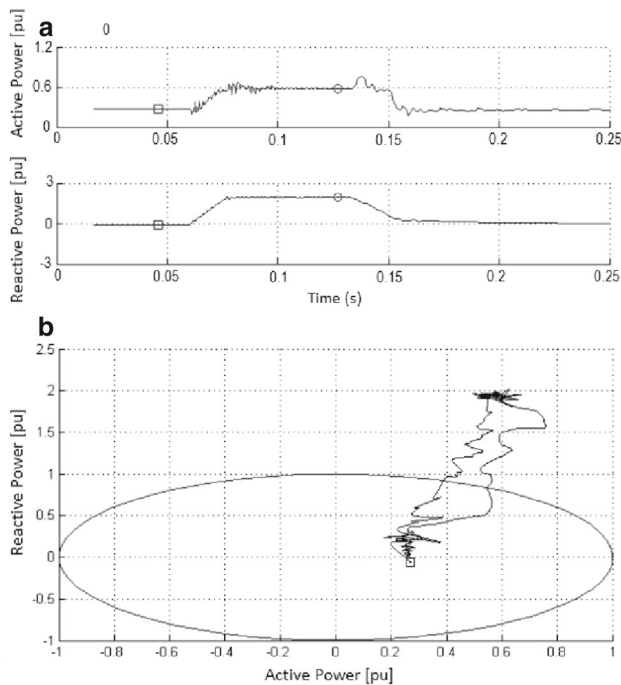


Fig. 4 **a** Power-flow behavior during the voltage sag; **b** power-flow behavior at complex chart

(only one side of system monitored is connected to power system generation); ($D1$) first Boolean variable that indicates a downstream fault. It represents the condition of a single source system (only one side of system monitored is connected to power system generation); (U , D) Boolean variables that indicate the location of the voltage sag source at the upstream or downstream system, respectively.

The proposed approach evaluated by a set of logics is presented as a good contribution which, through a effective solution from Boolean logics, indicates the VSS side. Moreover, the algorithm may be implemented on protective devices already installed with programmable Boolean logic blocks available and, as consequence, allowing the increase in the number of IEDs installed at interconnection points between utilities and consumers in order to indicate the VSS side.

3 Performance Evaluation: Simulated Cases Study

To validate the present proposal, data from voltage sag events obtained through the simulation of two systems were used. The transmission and distribution system and the cogeneration industrial system were simulated to provide a large possibilities of behavior when voltage sags occur. The simulations were performed using the Alternative Transient Program (ATP) software, and the evaluated algorithms were implemented in MATLAB software.

3.1 Transmission and Distribution System

The first simulated case study was based on actual data of a real Brazilian transmission and distribution system, according to the single-line diagram shown in Fig. 5. The system consists in long and noncompensated lines at 230 and 138 kV voltage levels, with large distances between load centers and power generation plants. The main generation units are natural gas, hydroelectric plants and sugarcane bagasse—fired thermoelectric plants. It has 93 transformers with 2,076 MVA installed capacity, 6,619 km of 230 and 138 kV transmission lines. This system has also a generation park formed by several concessionaires, self-producers and independent producers, with about 1,643 MVA of generation capacity.

The meters were installed in 7 points of the system (lower voltage side of power transformers), with 5 meters on the boundary between the 230 and 138 kV systems (M1, M2, M3, M4, and M5), 1 meter between the 138 and 13.8 kV system (M6) and 1 meter between the 230/230 kV system (M7), as shown in the diagram of Fig. 5. A total of 12 fault points were defined, 8 of them (F1, F2, F3, F4, F5, F9, F10 and F11) being located in the 138 kV system, 3 points (F6, F7 and F8) simulated in the 230 kV system and, lastly, 1 point (F12) located in the medium voltage system (13.8 kV). At each point were simulated two types of fault: three-phase and single-phase-to-ground with duration of 70 ms and incident angle of 0° . In all cases, eight fault resistance values were used: 0.001 (solid fault), 0.1, 1, 5, 10, 20, 40 and 80 Ω . Such combinations yielded 192 fault cases, evaluated in the 7 measurement points, totaling 1344 cases for testing the all algorithms. The result of each simulation is the indication of VSS side for those IEDs which detect a voltage sag. Table 2 presents the true voltage sag source location for each simulated fault, where (“u”) indicates upstream system and (“d”) indicates downstream system. The upstream and downstream system concepts were defined from the higher voltage level system to the lower voltage level system.

3.2 Cogeneration Industrial System

The second simulated case study was based on actual data of a Brazilian oil refinery, according to the single-line diagram shown in Fig. 6. The cogeneration industrial system is composed of five turbogenerators and approximately 70 MW of loads interconnected to the utility system by two 138 kV transmission lines (Oliveira Passos et al. 2014). Three of five turbogenerators are driven by steam turbines and two driven by gas turbines, totaling 80 MVA of generation capacity. As shown in Fig. 6, the generation and the distribution to the internal loads are made by eight buses of 13.8 kV, interconnected via insulated cables. It has five delta-wye transformers (Dy11), with 85 MVA installed capacity, connecting the 138 kV system to the 13.8 kV systems. The neutral resis-

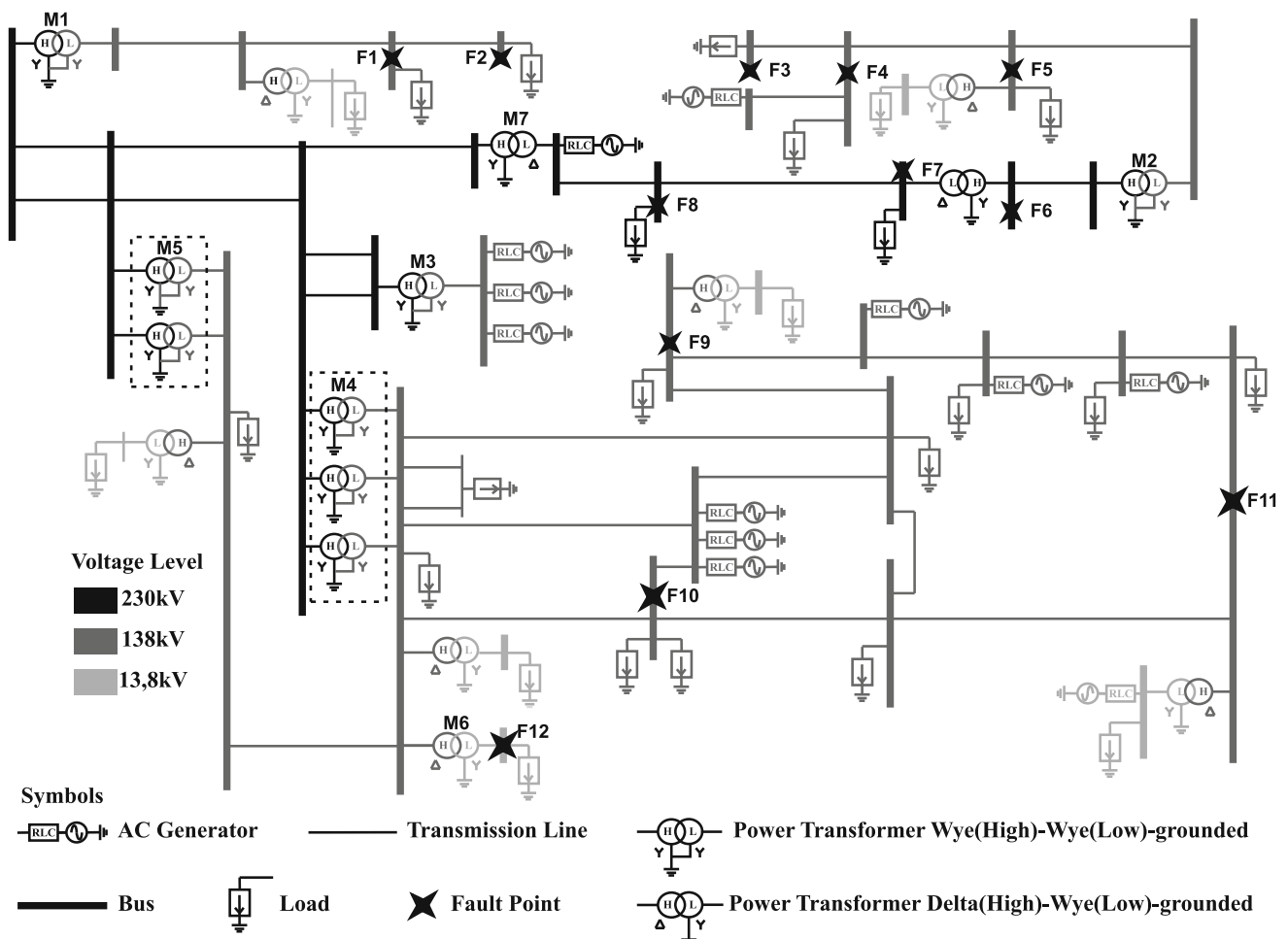


Fig. 5 Transmission and distribution system

Table 2 True sag source location—transmission and distribution system

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
M1	d	d	u	u	u	u	u	u	u	u	u	u
M2	u	u	d	d	d	u	u	u	u	u	u	u
M3	u	u	u	u	u	u	u	u	u	u	u	u
M4	u	u	u	u	u	u	u	u	d	d	d	d
M5	u	u	u	u	u	u	u	u	d	d	d	d
M6	u	u	u	u	u	u	u	u	u	u	u	d
M7	u	u	d	d	d	d	d	u	u	u	u	u

tors of 19Ω are installed on secondary winding of all power transformers. The loads are, typically, composed of 60 % of induction motors (constant power) and 40 % of other loads (constant impedance).

The meters were installed in 5 points of the system (lower voltage side of all power transformers), as shown in the diagram of Fig. 6. A total of 9 fault points were defined, 4 of them (F1, F2, F3 and F4) being located in the 138 kV system

and 5 point (F5, F6, F7, F8 and F9) located in the medium voltage system (13.8 kV). At each point were simulated two types of solid faults: three-phase symmetrical and single-phase-to-ground with duration of 70 ms and incident angle of 0° . Such combinations yielded 18 fault cases, evaluated in the 5 measurement points, totaling 90 cases for testing the all algorithms. The result of each simulation is the indication of VSS side for those IEDs which detect a voltage sag. Table 3 presents the true voltage sag source location for each simulated fault, where (“u”) indicates upstream system and (“d”) indicates downstream system.

3.3 Obtained Results

To evaluate the performance of the proposed method and the impedance method (Yonghai et al. 2012), as well as to compare them, the same simulated signals were used for both methods. It was found that the 192 faults simulated on the transmission and distribution system resulted in 504 voltage sags, representing an average of 2.6 voltage sag occurrences per fault. Table 4 shows the number of voltage sags detected

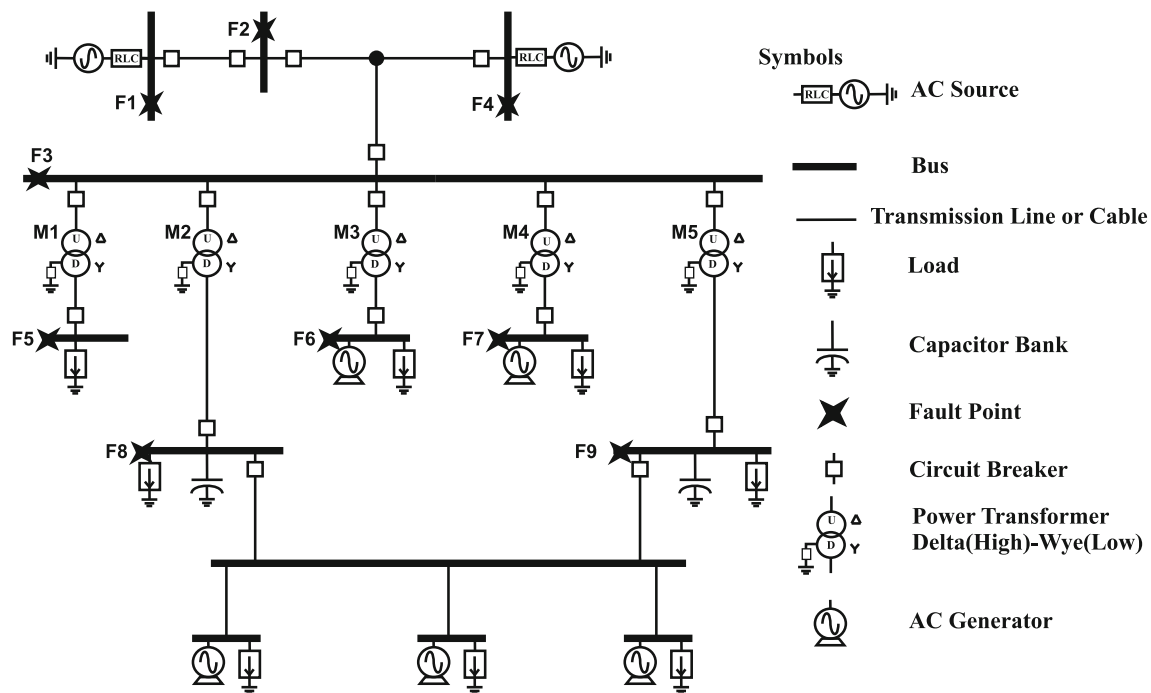


Fig. 6 Cogeneration industrial system

Table 3 True sag source location—cogeneration industrial system

	F1	F2	F3	F4	F5	F6	F7	F8	F9
M1	u	u	u	u	d	u	u	u	u
M2	u	u	u	u	u	u	u	d	d
M3	u	u	u	u	u	d	u	u	u
M4	u	u	u	u	u	u	d	u	u
M5	u	u	u	u	u	u	u	d	d

Table 4 Voltage sag occurrences for three-phase faults—transmission and distribution system

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
M1	8	8	0	0	0	0	0	0	0	6	0	0
M2	0	0	8	8	8	8	8	8	0	7	5	0
M3	0	0	0	0	0	0	0	8	0	7	6	0
M4	0	0	0	0	6	8	8	8	6	8	7	3
M5	0	0	0	0	0	0	0	8	0	7	5	0
M6	0	0	0	0	7	8	8	8	7	8	8	3
M7	0	0	0	8	8	8	8	8	0	7	6	0

by each IED when the eight different fault resistance values for three-phase fault type. Table 5 shows the number of voltage sags detected by each monitor when the eight different fault resistance values for single-phase-to-ground fault type.

Tables 6 and 7 present the ratio between the number of succeed VSS side indication and the number of voltage sags detected at each monitor for each fault considering the eight

Table 5 Voltage sag occurrences for single-phase-to-ground faults—transmission and distribution system

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
M1	8	8	0	0	0	0	0	0	0	5	0	0
M2	0	0	8	8	8	8	8	8	0	6	0	0
M3	0	0	0	0	0	0	0	0	0	7	0	0
M4	0	0	0	0	0	8	6	8	5	7	7	2
M5	0	0	0	0	0	0	0	0	0	6	0	0
M6	0	0	0	0	0	8	8	8	6	8	8	3
M7	0	0	0	0	7	8	8	8	0	6	0	0

Table 6 Ratio between the number of succeed VSS side indication and the number of voltage sags detected at each IED for each three-phase fault considering all different fault resistance values—transmission and distribution system

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
M1	1	1	1	1	1	1	1	1	1	1	1	1
M2	1	1	1	1	1	1	1	1	1	1	1	1
M3	1	1	1	1	1	1	1	1	1	1	1	1
M4	1	1	1	1	1	1	1	1	1	1	1	1
M5	1	1	1	1	1	1	1	1	1	0.86	0.8	1
M6	1	1	1	1	1	1	1	1	1	1	1	1
M7	1	1	1	1	1	1	1	1	1	1	1	1

different fault resistance values. Table 6 presents the results for all cases of three-phase fault (0.001, 0.1, 1, 5, 10, 20, 40 and 80 Ω), and Table 7 presents the results for all cases of single-phase-to-ground fault (0.001, 0.1, 1, 5, 10, 20, 40 and

Table 7 Ratio between the number of succeed VSS side indication and the number of voltage sags detected at each IED for each single-phase-to-ground fault considering all different fault resistance values—transmission and distribution system

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
M1	1	1	1	1	1	1	1	1	1	1	1	1
M2	1	1	1	1	1	1	1	1	1	1	1	1
M3	1	1	1	1	1	1	1	1	1	1	1	1
M4	1	1	1	1	1	1	1	1	1	1	1	1
M5	1	1	1	1	1	1	1	1	1	0.83	1	1
M6	1	1	1	1	1	1	1	1	1	1	1	1
M7	1	1	1	1	1	1	1	1	1	1	1	1

Table 8 Voltage sag results when three-phase fault—cogeneration industrial system

	F1	F2	F3	F4	F5	F6	F7	F8	F9
M1	1	1	1	1	1	0	0	0	0
M2	1	1	1	1	0	0	0	1	1
M3	1	1	1	1	0	1	0	0	0
M4	1	1	1	1	0	0	1	0	0
M5	1	1	1	1	0	0	0	1	1

Table 9 Voltage sag results when single-phase-to-ground fault—cogeneration industrial system

	F1	F2	F3	F4	F5	F6	F7	F8	F9
M1	1	1	1	1	1	0	0	0	0
M2	1	1	1	1	0	0	0	1	1
M3	1	1	1	1	0	1	0	0	0
M4	1	1	1	1	0	0	1	0	0
M5	1	1	1	1	0	0	0	1	1

80 Ω). The 18 faults simulated on the cogeneration industrial system resulted in 54 voltage sags, representing an average of 3.0 voltage sags occurrences per fault. Tables 8 and 9 present the voltage sag results for each case simulated. The “0” value represents no voltage sag, and “1” represents voltage sag occurrence.

3.3.1 Improved Impedance Method Results

In order to compare the improved impedance method to the proposed method, 504 voltage sags were recorded on the transmission and distribution system and 54 voltage sags were recorded on cogeneration industrial system; the algorithm correctly located the VSS side of the disturbance in 423 events and 45 events, respectively. These results represent a hit percentage of 83.9 and 83.3 %, respectively. The 90 errors

obtained were due to, mainly, the difficulty in determining an appropriate threshold between upstream and downstream cases at both simulated systems as shown in Figs. 7 and 8. The method proposed by Yonghai et al. (2012) considers -40% to the “delta impedance” and 25° to the “delta angle” as default.

3.3.2 Proposed Boolean Logic Method Results

From the same recorded voltage sags, the algorithm correctly located the VSS side of the disturbance in 501 events to the transmission and distribution system cases (99.4 %) and 54 events to the cogeneration industrial system (100.0 %). To the transmission and distribution system, only three events resulted a poor VSS side indication: M5 when F11 three-phase fault with fault resistance of 10 Ω ; M5 when F10 three-phase fault with fault resistance of 40 Ω ; and M5 when F10 single-phase-to-ground fault with fault resistance of 40 Ω . As matter of facts, the points of fault (F10 and F11) are electrically far from M5. As an example, the voltage sag recorded of 0.88 p.u. (shallow event) at M5 for three-phase fault with fault resistance of 10 Ω at F11, the three-phase apparent power variation from prevoltage sag condition to the voltage sag condition was only 2.5 %. This small variation recorded was not sufficient to ensure good performance of the method to this event.

Evaluating the two methods is notoriously the best performance of the proposed Boolean method, as shown in Fig. 9. Limitations identified in the improved impedance method were not observed in the proposed method, considering that no threshold is necessary, but only the power-flow behavior.

Voltage sags occur during short circuits, motor starting and transformer energizing (Lee et al. 2008). Almost all cases result in a voltage sag by the same reason. The higher power flow flowing through the system is mandatory to define the voltage sag, and the behavior of power flow during motor starting, transformer energizing, and short circuits are mostly the same. Therefore, the expected power-flow behaviors tracked by this proposed method are, mostly, the same for motor starting or transformer energizing.

Among the limitations that have been found in the proposed methods, stand out those cases where there are small variations in power before and during the occurrence of voltage sags, as could be found in shallow voltage sag occurrences. However, this limitation identified in other methods (Polajzer et al. 2015) was attenuated through this proposal, considering that the large number of hits to the higher resistance fault cases and the shallow voltage sags do not mostly cause problems to the systems (Pedra et al. 2005).

Fig. 7 Distance impedance method results—transmission and distribution system

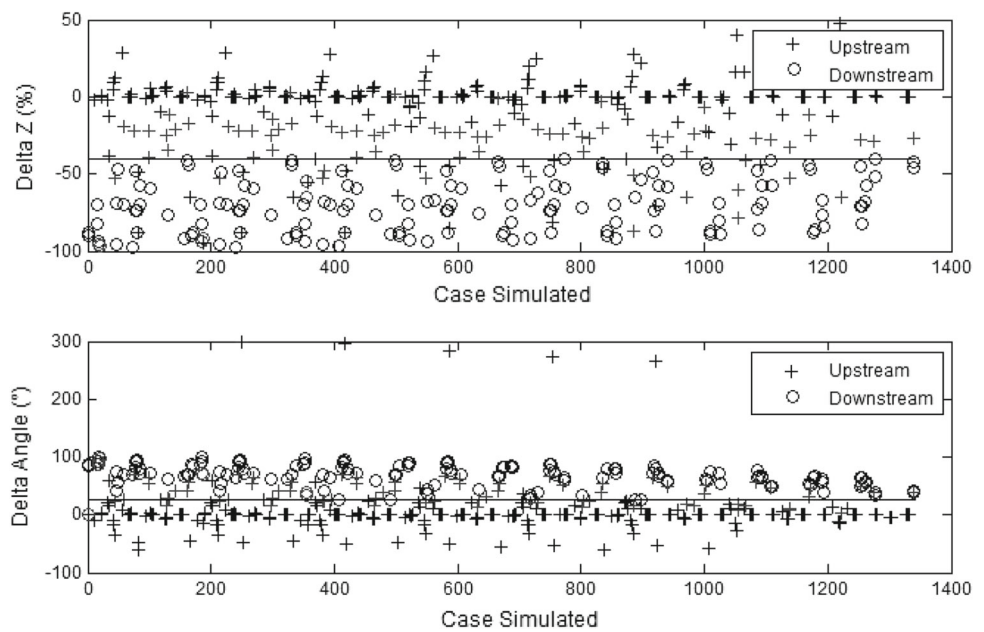
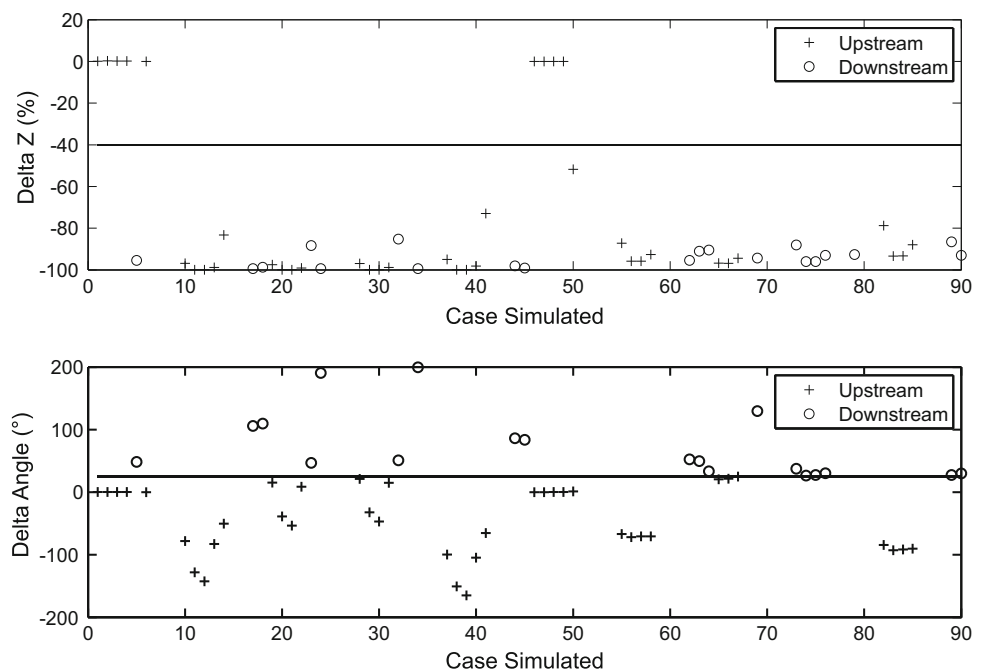


Fig. 8 Distance impedance method results—cogeneration industrial system



4 Conclusions

Several methods for VSS side indication based on power-flow measurement have been analyzed, characterized by their advantages and disadvantages. In this context, the new approach presented in this paper seeks to expand the application of the voltage sag tracking method. The algorithm based on Boolean logics bring an innovative way to indicate the VSS side, where the expected power flow behavior is verified. Active power sources at one side only or both sides, different

characteristics of loads (constant impedance, constant power or constant current) and the condition of operation at the time of the disturbance (1° , 2° , 3° or 4° quadrant) were contemplated and evaluated by this proposal. Thus, two systems with different characteristics have been simulated and the proposed algorithms presented an overall good performance with 99.5 % hits in locating the voltage sag source.

The authors intend to conduct complementary investigations and analyses: 1—to develop and test the prototype meter by hardware-in-the-loop using a Real-Time Digi-

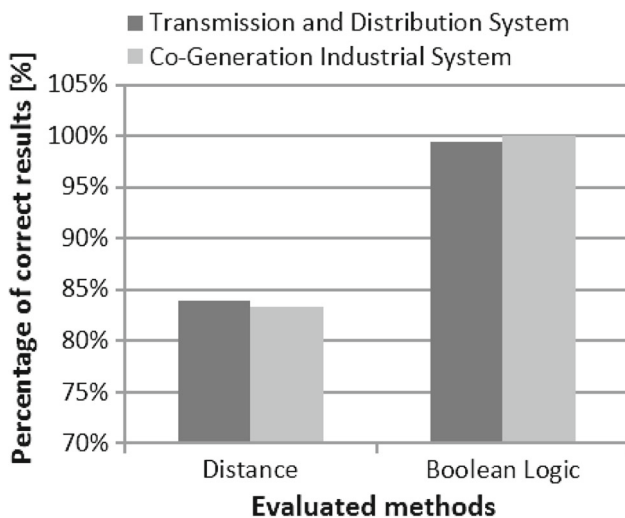


Fig. 9 Results summary

tal Simulator (RTDS); 2—to install a prototype meter in an actual system to confirm the numerical computation results.

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